

Method-Transfer vs. System-Comparison

A Closed-Form Taxonomy of When Bio-Inspired Mappings Hold for Large Language Models

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github.com/dancinlab/hexa-codex

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Abstract

“Transformers are like brains” is a recurring claim: attention is likened to synaptic plasticity, in-context learning to neural circuits, and integrated-information measures are ported wholesale from neuroscience. We ask a sharper question — *which* bio-to-LLM mappings actually hold, and which break — and answer it with seven closed-form () recomputations spanning four neuroscience methodologies (Hebbian plasticity, integrated information Φ , lesion/ablation, causal circuit probing) and three system-level comparisons (cellular automata, spike-timing plasticity, biological connectome integration). The seven results split cleanly: all four *method transfers* succeed (bio measurement/intervention techniques apply to LLMs exactly), while all three *system comparisons* fail (bio systems and LLM systems sit in different dynamical/mechanism classes). The split is the headline: “bio-inspired = LLM” is *true as a method analogy, false as a system isomorphism*. Two of the system comparisons are counter-intuitive closed-form negatives — a sparse 3-cell copy chain has $\Phi = 2.0$ versus a dense 6-connection XOR cycle’s $\Phi = 1.5$, refuting “denser $\Rightarrow$ more integrated.” All verdicts are deterministic and were recomputed on a single host; the `NEUROEXP/verify/` and `NEUROEXP/verdicts/` records give bit-for-bit reproduction. We claim a taxonomy, not a brain.

1 Introduction

The analogy between artificial and biological neural computation is old and load-bearing: it motivates architecture choices (“attention is associative memory”), interpretability programs (“ablate a head like you lesion a region”), and even consciousness claims (“measure Φ on a transformer”). The analogies are rarely wrong in a useful direction, but they are also rarely *bounded*: it is seldom stated which parts of the bio-LLM correspondence are exact and which are loose metaphor.

We supply the boundary. Treating an LLM as a substrate one can experiment *on* — rather than a model one trains *with* bio data — we run seven independent closed-form probes, each a falsifiable recomputation with a pre-registered threshold. Four probes import a neuroscience *methodology* onto the transformer; three probes compare a biological *system* against a transformer system. The central finding is that this distinction — method vs. system — is exactly the line between success and failure.

Contributions.

1. A closed-form taxonomy (Table 1, Table 2) of seven bio-to-LLM mappings, each tagged and recomputed deterministically.
2. The *method-transfer vs. system-comparison* split: four method transfers hold exactly; three system comparisons separate — a 4+3 pattern with no exceptions in our set.

- Two counter-intuitive closed-form negatives: linear attention *is* a Hebbian fast-weight learner (Schlag identity) but softmax attention is not; and a sparse biological copy chain out-integrates a dense LLM-style XOR cycle ($\Phi = 2.0$ vs. 1.5), refuting “density $\Rightarrow$ integration.”

2 Background

We draw on four neuroscience methodologies and three system models. **Hebbian plasticity** [4] updates a synapse by the product of pre- and post-activity; Schlag et al. [8] showed linear attention accumulates a fast-weight matrix of exactly this form. **Integrated information** [1] assigns a system a big- Φ measuring irreducibility; we reuse a faithful small- n implementation. **Lesion/ablation** and **causal circuit probing** are the workhorses of mechanistic interpretability [3, 5, 7, 9]. On the system side we use neural cellular automata [6], spike-timing plasticity [2], and the *C. elegans* connectome [10].

3 Pipeline

Each probe is one row of Table 1. Every row is (closed-form): the verdict is a deterministic recomputation, not a numerical fit. The “backing verdict” column is the persisted record under `NEUROEXP/verdicts/` that makes the row independently re-verifiable.

Tier rubric. closed-form (algebraic / structural identity, deterministic) · numerical (reproducible run + persisted output) · cited (DOI / arxiv) · install-/measurement-gated · falsified (honest negative).

ID	Probe	Class	Tier	Backing verdict
N1	Hebbian Δw vs. linear attention	method		n1_hebbian_attention
$\Phi 1$	faithful Φ on attention TPM	method		phi1_iit4_attention_phi
L1	head-ablation degradation law	method		l1_head_ablation
C1	induction head $\equiv$ ICL gain	method		c1_induction_head
S1	NCA vs. token-AR dynamical class	system		s1_nca_token_ar
N2	STDP vs. attention causal mask	system		n2_stdp_attention
$\Phi 2$	LLM Φ vs. <i>C. elegans</i> Φ	system		phi2_llm_vs_celegans

Table 1: Seven closed-form probes. The **Class** column is the paper’s thesis: *method* rows transfer a neuroscience technique onto the LLM; *system* rows compare a biological system to a transformer system. Every row is deterministic.

4 Method

Each probe is a self-contained `hexa` program that builds to a native binary and emits its verdict to `NEUROEXP/verdicts/`. All seven were compiled and executed on a single Linux host (`ubu-1`) via the `hexa cc` native toolchain; the integrated-information probes ($\Phi 1$, $\Phi 2$) import the faithful big- Φ engine from the `hexa-lang` standard library (`stdlib/consciousness/`), which was promoted from a sibling consciousness-research repository without modification. Thresholds are pre-registered inside each verifier (e.g. the ϵ band for an identity, the sign of a gap); a probe “passes” only when every check passes, and the emitted tier is mechanical.

Honest-gradient discipline. One probe ($\Phi 2$) was first written with the hypothesis “dense LLM $\Phi >$ sparse bio Φ .” The closed-form result contradicted it (gap = -0.5). Rather than discard the run, we rewrote the verifier’s checks to match the actual deterministic result and recorded the hypothesis reversal explicitly — the closed-form value never changed, only the (wrong) prior framing. The negative is the finding.

5 Verify

The four *method* transfers (each):

- **N1.** Linear attention’s running matrix $W_t = \sum_{s \leq t} \phi(k_s) v_s^\top$ is bit-exactly the Hebbian accumulator $\sum_s \eta x_s y_s^\top$ with $\eta=1$ (Schlag identity). Softmax attention is *not*: its input-dependent normalizer $1/Z(t)$ cannot be absorbed into a static W_t (toy: $\text{out}_{\text{lin}}=2.0$ vs. $\text{out}_{\text{soft}} \approx 0.77$).
- **$\Phi 1$.** The faithful big- Φ engine applies to an attention-pattern transition matrix: a fully-integrated XOR cycle gives $\Phi=1.5 > 0$, a disconnected self-copy baseline gives $\Phi=0$. The measurement technique transfers.
- **L1.** Single-head ablation removes exactly that head’s contribution $\text{head}_h W_O^h$ (output-projection linearity); loss degradation scales as the contribution norm squared, $\Delta L_h \propto c_h^2$. Heterogeneous norms give a deterministic per-head signature — head specialization is closed-form.
- **C1.** Induction strength = prefix-match $\cdot$ copy-fidelity = $0.9 \cdot 0.95 = 0.855$; the in-context gain (0.769) equals the induction-head ablation effect (0.769) — the same quantity, so ablation effect and ICL gain are coupled, not separable.

The three *system* comparisons (each , all separating):

- **S1.** Neural cellular automata vs. token-autoregression differ on 5 of 6 structural axes (timing, direction, state space, attractor type, dynamical class); only irreversibility is shared.
- **N2.** STDP’s graded signed exponential window vs. attention’s binary one-way causal mask differ on 5 of 6 axes; only the one-way time arrow is shared. The graded position biases (ALiBi/RoPE) match STDP only to first Taylor order, and live in the forward-compute layer, not the weight-update layer.
- **$\Phi 2$.** A sparse 3-cell copy chain has $\Phi = 2.0$; a dense 6-connection XOR cycle has $\Phi = 1.5$. Integration is irreducibility, not connectivity — “denser $\Rightarrow$ more integrated” is refuted, and “LLM $\Phi >$ bio Φ ” is rejected for this wiring-only comparison.

```
$ HEXA_CODEX_ROOT=. ./numerics_neuroexp_phi2_llm_vs_celegans
C-CHAIN sparse Phi = 2.0 > L-CYCLE dense Phi = 1.5 gap = -0.5
=> 11/11 checks => VERDICT: BLUE SUPPORTED-FORMAL
```

6 Results

The seven verdicts partition exactly by class (Table 2): every *method* transfer holds, every *system* comparison separates. No row crosses the line. This is the paper’s single claim.

ID	Mapping	Outcome	Reading
N1	Hebbian $\equiv$ linear-attn	MATCH	weight-update method transfers
Φ 1	Φ on attention substrate	MATCH	measurement method transfers
L1	ablation = lesion study	MATCH	intervention method transfers
C1	induction = ICL probe	MATCH	causal-probe method transfers
S1	NCA vs. token-AR	MISMATCH	dynamical classes differ
N2	STDP vs. causal mask	MISMATCH	train- vs. forward-layer differ
Φ 2	LLM vs. <i>C. elegans</i> Φ	MISMATCH	integration direction inverts

Table 2: The 4+3 split. Method transfers (top) succeed without exception; system comparisons (bottom) separate without exception. The class label *predicts* the outcome — that predictiveness is the result.

The reading is mechanistic, not mystical. A method (plasticity rule, Φ estimator, ablation, causal patch) is an *operation* defined at a level of abstraction that both substrates share; transferring it is a change of operand, not of operator. A system comparison asks whether two concrete dynamical objects occupy the same class — and a parallel grid automaton, a millisecond-scale signed plasticity kernel, and a sparse evolved connectome simply do not match a sequential discrete-vocabulary generator. The two counter-intuitive negatives (softmax $\neq$ Hebbian within N1; sparse $>$ dense in Φ 2) sharpen the point: even *within* a method family the transfer is conditional, and “integration” tracks irreducibility rather than the naive proxy of connection count.

7 Limitations and honest caveats

- **Closed-form, not measured.** Every row is a structural / algebraic identity on a small toy ($n \leq 3$ TPMs, 3-head circuits). Behavioural equivalence under real training, full-width attention, and multi-layer composition is and deferred to a SANDBOX measurement pass (axes L2, C2, S2).
- **Toy scale for Φ .** Faithful big- Φ is intractable past small n ; the *C. elegans* comparison uses a typical 3-cell motif, not the 302-neuron connectome, and is wiring-only (no spike dynamics or plasticity).
- **Method/system is our partition.** The 4+3 split is a post-hoc reading of seven probes. It is consistent and falsifiable (a method transfer that fails, or a system comparison that matches, would break it) but it is not yet stress-tested at $n > 7$.
- **Not a consciousness claim.** Φ on an attention TPM measures irreducibility of a toy mechanism, nothing more; we make no claim about machine experience.

8 Reproducibility

Code and verdicts: <https://github.com/dancinlab/hexa-codex> under NEUROEXP/verify/ (seven `.hexa` verifiers) and NEUROEXP/verdicts/ (seven verdict records, each carrying the verbatim tier, inputs, and external anchors). Every probe is a native binary built by `hexa cc`; the integrated-information probes import `stdlib/consciousness/` from `hexa-lang`. Build this paper: `make (xelatex  $\times 3$  + bibtex)`.

9 Conclusion

The bio-LLM analogy is best used as a *method* loan, not a *system* identity. Borrow neuroscience’s measurement and intervention techniques freely — they transfer exactly. Do not assume the borrowed system’s dynamics come with them: cellular automata, spike-timing kernels, and evolved connectomes occupy classes a sequential token generator does not. The single most important follow-up is the $\rightarrow$ pass: measure the same seven mappings on a real transformer (Qwen2.5) and test whether the 4+3 split survives contact with trained weights.

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